The worst thesis in the world

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ABSTRACT

[This abstract must provide a succinct and informative condensation of your work. Candidates are welcome to prepare a lengthier abstract for inclusion in the dissertation, and provide a shorter one in the CaltechTHESIS record.]

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First author contributions: [1], [2]

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Chapter 1

INTRODUCTION

The gravity that we know here on Earth is intuitive. It is familiar. It is expected. And it is safe (for the most part). But this universe that we live in continues to offer so much more than we expected. Starting from observing the motion of the planets and stars in the night sky to theorizing about black holes, the most compact and extreme objects in the universe, gravity continues to humble us.

1.1 Newtonian Gravity

In order to explain black holes, the overarching topic of this thesis, we must first start with a more simplified version of gravity and work our way up. Newtonian gravity states that the gravitational force between two masses is proportional to the two masses and inversely proportional to the square of the distance between the masses, or

$$\mathbf{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}}, \tag{1.1}$$

with G being Newtons constant. However, we usually don't just have point masses like m_1 and m_2. We often have a distribution of matter which we'll represent as a matter density ρ. This matter distribution will generate a gravitational potential Φ via Poisson's equation

$$\nabla^2\Phi = 4\pi G\rho. \tag{1.2}$$

To relate Eqs. 1.1 and 1.2, we compute the gradient of Φ as a force,

$$\mathbf{F} = -\nabla\Phi. \tag{1.3}$$

Then, because of Newtons generic force equations

$$\mathbf{F} = m\mathbf{a} = m\frac{d^2\mathbf{x}}{dt^2}, \tag{1.4}$$

we can get the equations of motion for any object in Newtonian gravity,

$$m\frac{d^2x_i}{dt^2} = -\frac{\partial\Phi}{\partial x_i}. \tag{1.5}$$

Additionally, there is a physical constraint on the matter density ρ, namely the conservation of energy. This looks like

$$\frac{\partial\rho}{\partial t} = -\nabla\cdot(\rho\mathbf{v}). \tag{1.6}$$

This equation states that the rate of change of matter over time is given by the flux of matter entering or leaving a region.

With these equations, one is able to compute quite a lot of gravitational scenarios. This could be the motion of a planet around a star or moving a satellite to a Lagrange point of two orbiting bodies. Combined with some ideal fluid mechanics for computing ρ, they can also be used to model the interiors of large gas planets or even stars. These basic starting equations are able to unlock a large amount of gravitational physics on their own, making them vital, even today, to astrophysical research.

However, Newtonian gravity does have its limitations. One being how simultaneity of events is measured. In order to understand simultaneous events, we must understand the coordinates that are used and how they transform in Newtonian gravity. If we take a step back, one way to conceptually think about Newtonian gravity is that we have a time coordinate t. And at each time t, we have a snapshot of our 3D world which is represented by some spatial coordinates x, y, z. Therefore, two events that happen simultaneously will be labeled with the same value for the time coordinate t, no matter where we are in our spatial coordinates. However, we have some freedom with our spatial coordinates. We can freely transform them from cartesian (x, y, z) to spherical (r, θ, ϕ) to cylindrical (s, ϕ, z) or to any other coordinates we can think of without altering our time coordinate, preserving simultaneity. Conversely, the only way to transform our time coordinate (and keep the same physics) is a simple $t \to t + \delta t$. Therefore, time and space are somewhat decoupled in Newtonian gravity.

Another limitation of Newtonian gravity is that it doesn't account for the finite speed of light. Related to the first limitation, events can only be considered simultaneous if they occur at the same time t. Consider a scenario where one person stands at rest at a point (x, y, z) and time t and shines a flashlight in the $+x$ direction. A second person is moving with respect to the first person in the $+x$ direction with a velocity v. This second person has their own coordinate system (t', x', y', z') different from the first person at rest and would find the relation between the two systems to be

$$t' = t \tag{1.7}$$

$$x' = x - vt \tag{1.8}$$

$$y' = y \tag{1.9}$$

$$z' = z. \tag{1.10}$$

Consider what happens if the second person is moving very close to the speed of light (i.e. $v \approx c$). One may be tempted to think of this scenario as similar to a different scenario where the first person is throwing a baseball while the second rides in a car. It'd be perfectly reasonable for a car to be moving faster than a baseball. However, this is fundamentally impossible with light. The speed of light is constant in all coordinate systems. So if we try to interpret this scenario in a Newtonian way, the second person would be moving almost in line with the front of the light beam at the same speed. But then what would happen if this second person shined their own flashlight? Would the this second beam of light be moving almost twice as fast as the first one? Would it stay effectively stationary and not go anywhere because the second person was already moving at the speed of light? This inconsistency and impossibility is why Newtonian gravity isn't the full picture and why we need a different theory.

1.2 Special Relativity

The theory of special relativity (SR) is able to reconcile and account for the finite speed of light in all coordinate systems. Say we have some event P. Fig. 1.1 shows a light cone centered at event P. The vertical axis is time and the horizontal axis is a spatial dimension. The lines at $\pm 45°$ represent the paths that light take both into the future and from the past. In the "Past" light cone, everything contained within the light cone is able to influence event P in a causal way (i.e. things traveling at or slower than the speed of light). Everything within the "Future" light cone can be influenced by event P. Events within these light cones are said to be causally connected to P. However, everything outside the light cone is unable to affect or be influence by event P, thus we call those events causally disconnected.

In Newtonian gravity, we have a simple notion of an invariant distance between two points, e.g.

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \tag{1.11}$$

This measure of distance is independent of what coordinate system we use for our spatial coordinates and works for Newtonian gravity because all time coordinates of all observers are the same. There is a universal time in Newtonian gravity. However, in SR we no longer have this universal time which everybody can agree on. Therefore, the measure of "distance" that we use in SR must account for our time coordinate somehow because of the coupling between space and time that the finite speed of light demands. For now, we just present this distance measure, called

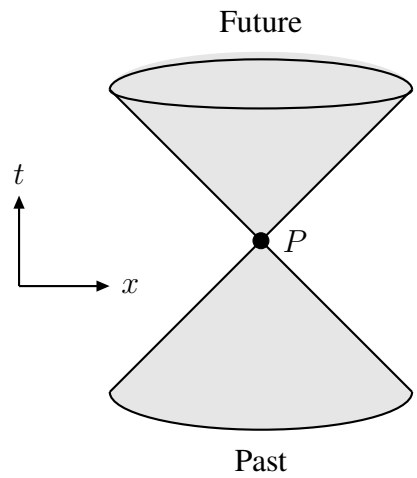

Figure 1.1: A light cone centered at the event P. The cone edges represent rays propagating at the speed of light c.

the spacetime interval, as

$$(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2. \tag{1.12}$$

First, notice that the units of this don't match. We are adding something like (seconds)^2 to something like (meters)^2. But one can notice that if we multiply the Δt by the speed of light c, we'll get the proper units. Very quickly in relativity, it becomes annoying to always have to write and keep track of this factor of c. Therefore, almost everybody in the field simply sets $c = 1$ with the assumption that if we ever need to calculate actual quantities, we'll add it back in at the end.

Second, we will now use this spacetime interval to motivate constructing something called a spacetime "metric". Now to fully understand the significance of what a metric is and what it represents requires taking courses on differential geometry. However, for our purposes, a metric is basically a 4×4 matrix that represents how coordinates combine together to form the spacetime interval. Therefore, for the interval in Eq. 1.12, we can define its spacetime metric, named the Minkowski metric, as

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.13}$$

It is diagonal because there are no cross-terms in Eq. 1.12 (such as $\Delta x \Delta y$). We can then rewrite Eq. 1.12 using Einstein summation notation (where all repeated

indices are implicitly summed)

$$(\Delta s)^2 = \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu \tag{1.14}$$

where

$$x^\mu = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}. \tag{1.15}$$

With a spacetime metric, we are able to do so many things. We can raise and lower indices of different quantities (i.e. transform between vectors and dual vectors/one-forms, $x_\nu = \eta_{\mu\nu}x^\mu$). We can turn the spacetime interval from Δs into infinitesimals and integrate it to find the length of a spacetime curve. But one of the most powerful things we can do is define transformations between coordinates that leave spacetime intervals unchanged (similar to Newtonian gravity). Lets define a transformation from an un-primed coordinate system to a primed one with

$$x^{\mu'} = \Lambda^{\mu'}{}_{\mu} x^\mu. \tag{1.16}$$

To show that this leaves the spacetime interval unchanged, let's plug it in

$$\begin{aligned}(\Delta s)^2 &= \eta_{\mu\nu}\Delta x^\mu \Delta x^\nu \\ &= \eta_{\mu'\nu'}\Delta x^{\mu'}\Delta x^{\nu'} \\ &= \eta_{\mu'\nu'}\Lambda^{\mu'}{}_{\mu}\Delta x^\mu \Lambda^{\nu'}{}_{\nu}\Delta x^\nu \\ &= \eta_{\mu'\nu'}\Lambda^{\mu'}{}_{\mu}\Lambda^{\nu'}{}_{\nu}\Delta x^\mu \Delta x^\nu.\end{aligned} \tag{1.17}$$

Therefore, Λ is a transformation between coordinates only if

$$\eta_{\mu\nu} = \eta_{\mu'\nu'}\Lambda^{\mu'}{}_{\mu}\Lambda^{\nu'}{}_{\nu}, \tag{1.18}$$

Or said another way, Λ doesn't transform the components of the metric. If Λ satisfies this, then we call it a Lorentz transformation. These will be mentioned in Ch. **??**.

1.3 General Relativity

Now in going from Newtonian gravity to SR, we didn't actually add in any gravity to SR. We only tackled the issues of simultaneity and the speed of light. Now we are ready to combine SR with gravity to produce Einstein's theory of general relativity (GR). Probably the biggest conceptual difference between GR and Newtonian gravity is that Newtonian gravity is explained as a force exerted between two masses. And

any object that has a force being applied to it will accelerate in the direction of that force. Therefore, motion in Newtonian gravity arises from forces.

In GR, however, motion does not arise from forces. Instead, in GR we say there is some "background" spacetime, or some fabric of the universe that matter exists both on and in at the same time. In the Newtonian and SR picture, this background is "flat" meaning that it doesn't affect the matter that lives in it. However, in GR this background doesn't have to be flat anymore. It can be distorted, or curved; sort of how a trampoline with a heavy object in the middle is curved. Motion then arises not because of any force, but simply due to the curvature of the background spacetime. The motion of a small mass is dictated by how much spacetime is curved. In the trampoline example, a small mass starting with a constant velocity on a flat spacetime (flat trampoline) will not deviate from its original path and will continue with a constant velocity. That same mass moving on a curved spacetime (trampoline with a heavy object in the middle) will naturally move towards the heavy object with increasing velocity without any outside influence. It's the nature of the curved spacetime which drives the motion of matter. And the beauty of Einstein's theory is that it is matter itself which drives the curvature of spacetime.

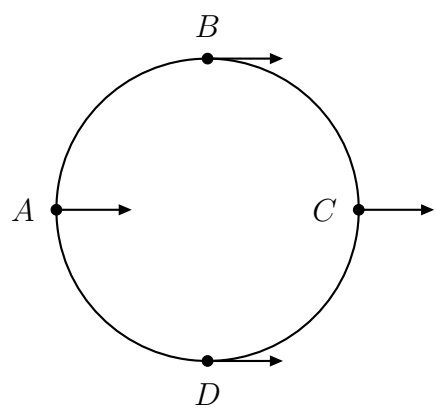

Figure 1.2: Diagram for parallel transporting a vector along two different paths of a flat circle: $A \to B \to C$ and $A \to D \to C$.

Therefore in GR, in order to describe gravity, we actually must describe the geometry of spacetime as this is what we experience as gravity. The first thing we should probably understand is how motion a curved spacetime is different from a flat spacetime. Let's demonstrate this by moving some vectors along some curves. In Fig. 1.2, we take a vector pointing to the right at point A and move it to point C along two different paths; $A \to B \to C$ and $A \to D \to C$. We move the vectors in such a way that doesn't rotate or scale them. Therefore, somewhat intuitively,

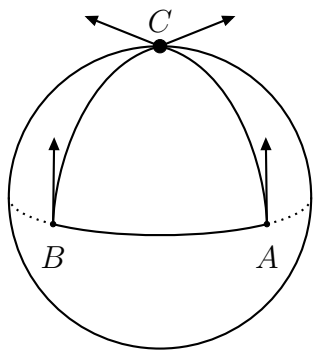

Figure 1.3: Diagram for parallel transporting a vector along two different paths of a sphere: $A \to B \to C$ and $A \to C$.

we expect both vectors at C to be identical because the vectors weren't changed in any way except for their positions. Now we do the same thing in Fig. 1.3, except this time we start with a vector at A that lies on the equatorial line of a sphere and move it to point C at the pole along two different paths; $A \to C$ and $A \to B \to C$. Along the first path, the vector will be pointing tangent to the pole towards the left of the image. Along the second path, the arrow is first moved along the equatorial line to B, and then moved to the pole pointing towards the right. Unlike in Fig. 1.2, the final vectors are not the same. This demonstrates the unintuitive nature of a curved space compared to a flat one, and why will we need some more machinery to explain GR.

The reason we are concerned with how vectors change as we move through our space is because any coordinate system that we create is made up of basis vectors. In a flat space, no matter what closed-loop path we choose, our basis vectors will always be the same at the same point. As shown in Fig. 1.3, this is not true for a curved space. Therefore, we must be careful about how we move vectors around our spacetime. To describe the curvature, similar to SR, GR will also have a metric associated with it. We will denote this metric as $g_{\mu\nu}$ because $\eta_{\mu\nu}$ is reserved for Minkowski spacetime. Our new metric $g_{\mu\nu}$ will encode everything we need to know about the gravity and curvature of our spacetime.

We start by expanding our notion of what a derivative is in our curved spacetime. Instead of the typical partial derivative $\partial/\partial x$, which works in a flat spacetime, we need a derivative works in a curved spacetime, will be independent of our coordinates, and reduces back to the partial derivative when the spacetime is flat. Since the partial derivative is already a good place to start, we can say that our new derivative

is the partial derivative plus some correction that accounts for the coordinates. We define the covariant derivative operator ∇_μ acting on the components of a vector V^λ as

$$\nabla_\mu V^\lambda = \partial_\mu V^\lambda + \Gamma^\lambda_{\mu\nu} V^\nu \tag{1.19}$$

where we have again made use of the Einstein summation notation of implicit sums over repeated indices. Here we call $\Gamma^\lambda_{\mu\nu}$ the connection coefficients as they connect basis vectors from different points in our spacetime. These connection coefficients are a general property of differential geometry and depend on the metric chosen, but in GR we call them the Christoffel symbols.

Before we define the Christoffel symbols in terms of the metric $g_{\mu\nu}$, we will impose two conditions that turn out to be necessary for GR. First is that our spacetime (and by extension the metric and the connection coefficients) are torsion free, meaning

$$\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}. \tag{1.20}$$

The second condition is that our connection is "metric compatible", meaning

$$\nabla_\rho g_{\mu\nu} = 0. \tag{1.21}$$

This second condition is important for raising and lowering indices of derivative quantities. With these two conditions, we can write the Christoffel symbols as sums of partial derivatives of the metric

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} \right). \tag{1.22}$$

Next we will define how to transport a vector in the most "constant" way possible along a path in a curved spacetime. This is useful because still want to be able to move vectors in our spacetime without them being rotated or changed in some way. We define parallel transport of a vector V^σ along a path $x(\lambda)$ in our spacetime

$$\frac{dx^\mu}{d\lambda} \nabla_\mu V^\sigma = \frac{dx^\mu}{d\lambda} \frac{\partial V^\sigma}{\partial x^\mu} + \Gamma^\sigma_{\mu\nu} \frac{dx^\mu}{d\lambda} V^\nu = 0 \tag{1.23}$$

If the vector that we apply parallel transport to is the tangent vector itself, $dx^\mu/d\lambda$, then we get the equation

$$\frac{d^2 x^\mu}{d^2\lambda} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0, \tag{1.24}$$

which is called the geodesic equation. The geodesic equation will be our notion of a "straight line" in a curved spacetime, i.e. a path without any external forces or

influence. If you squint hard enough, you may be able to see that this is the GR extension of acceleration $\mathbf{a} = d^2\mathbf{x}/dt^2 = 0$ in Newtonian gravity.

Now that we are able to compute how to move vectors around our spacetime, we can finally move on to computing the curvature of our spacetime which is where gravity in GR derives from. To do this, we start with a vector V^ρ. We already know if we move this vector along two different paths to the same destination, the resulting vectors will be different. This difference is precisely what we are going to call curvature. Imagine we first transport the vector in some direction ∇_ν and then in another direction ∇_μ. This gives us a final vector of $\nabla_\mu\nabla_\nu V^\rho$. Now, we swap the order to get $\nabla_\nu\nabla_\mu V^\rho$. Let's take the difference and expand the covariant derivatives

$$\begin{aligned}\nabla_\mu\nabla_\nu V^\rho - \nabla_\nu\nabla_\mu V^\rho &= \partial_\mu(\nabla_\nu V^\rho) - \Gamma^\lambda_{\mu\nu}\nabla_\lambda V^\rho + \Gamma^\rho_{\mu\sigma}\nabla_\nu V^\sigma - (\mu \leftrightarrow \nu) \\ &= \partial_\mu\partial_\nu V^\rho + \partial_\mu(\Gamma^\rho_{\nu\sigma})V^\sigma + \Gamma^\rho_{\nu\sigma}\partial_\mu V^\sigma - \Gamma^\lambda_{\mu\nu}\partial_\lambda V^\rho \\ &\quad - \Gamma^\lambda_{\mu\nu}\Gamma^\rho_{\lambda\sigma}V^\sigma + \Gamma^\rho_{\mu\sigma}\partial_\nu V^\sigma + \Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\nu\lambda}V^\lambda - (\mu \leftrightarrow \nu) \\ &= (\partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma})V^\sigma \end{aligned} \tag{1.25}$$

In the last line we simplified quite a bit, especially using the torsion-free condition, but now we have the difference of moving a vector along two different paths. Let's rewrite this in a more compact manner

$$\nabla_\mu\nabla_\nu V^\rho - \nabla_\nu\nabla_\mu V^\rho = R^\rho{}_{\sigma\mu\nu}V^\sigma \tag{1.26}$$

where we call $R^\rho{}_{\sigma\mu\nu}$ the Riemann curvature tensor. To write it out more explicitly,

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{1.27}$$

The Riemann curvature tensor is made up of first and second derivatives of the metric because the Christoffel symbols (Eq. 1.22) have first derivatives of the metric in them. The Riemann tensor encodes everything we need to know about the curvature of our spacetime, and it is only computed from derivatives of our metric $g_{\mu\nu}$. There are a number of symmetries of the Riemann tensor which we won't go over, but they reduce the number of independent components from $4^4 = 256$ to just 20.

A useful contraction of the Riemann tensor is called the Ricci tensor

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}. \tag{1.28}$$

The Ricci tensor has the property of being symmetric, $R_{\mu\nu} = R_{\nu\mu}$, and only has 10 independent components. Again, we can contract the Ricci tensor to get the Ricci scalar

$$R = R^\mu{}_\mu. \tag{1.29}$$

Finally, we are able to combine our work to form the Einstein tensor

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}, \tag{1.30}$$

which has the nice property of $\nabla^{\mu}G_{\mu\nu} = 0$.

Now that we have the Einstein tensor, let us take a step back. In the Einstein tensor, we have sums of the metric $g_{\mu\nu}$, its first derivative squared $(\partial_{\rho}g_{\mu\nu})^2$, and second derivatives $\partial_{\lambda}\partial_{\rho}g_{\mu\nu}$. We know that all these derivatives somehow represent the curvature of our spacetime which in turn gives us gravity. Let's relate this back to Newtonian gravity to see if we can gain a little intuition. Recall Poisson's equation for a gravitational potential Φ (rewritten here for ease)

$$\nabla^2\Phi = 4\pi G\rho. \tag{1.31}$$

Φ encodes the information of our gravitational field. On the left, we have two derivatives of this gravitational potential and on the right we have the density of matter ρ. We also know that Newtonian gravity is a good approximation for GR in the case of weak gravity like we have here on Earth. We happen to have just constructed a quantity that contains second derivatives of the metric $g_{\mu}\nu$ which is our GR analog of the gravitational potential Φ.

Therefore, we can postulate, just as Einstein did, that the equations of GR have the following form:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}. \tag{1.32}$$

These are the infamous Einstein field equations. They are a set of 10 coupled non-linear second order partial differential equations (PDEs) for the spacetime metric. On the right, we have introduced a quantity $T_{\mu\nu}$ which is called the stress energy tensor. This tensor contains all the information about the matter and energy distribution in our spacetime. Because we have that $\nabla^{\mu}G_{\mu\nu} = 0$, we also have that $\nabla^{\mu}T_{\mu\nu} = 0$. Based on how we construct $T_{\mu\nu}$, this actually works out to be the conditions for conservation of energy and momentum, all wrapped up into one tensor equation. Also, it should be noted that G is Newtons gravitational constant, and not the trace of $G_{\mu\nu}$. We often set $G = 1$ as well to avoid confusion and make the equations easier to work with.

There is considerable physics to be studied when we include the stress energy tensor in the Einstein field equations such as neutron starts, white dwarf stars, cosmology, active galactic nuclei, black hole accretion disks, supernovae, and several others.

However, for this thesis we will be focusing on black holes. And even though many black holes form due to the collapse and/or accretion of matter, once a black hole is formed there need not be any matter outside the black hole to describe a black hole spacetime. Therefore, for the majority of this thesis, we will set $T_{\mu\nu} = 0$. This reduces the Einstein field equations significantly to

$$R_{\mu\nu} = 0. \tag{1.33}$$

As an example, let us look at the simplest black hole spacetime that exists; the Schwarzschild spacetime. This is a static spherically symmetric spacetime. The metric in spherical coordinates for the Schwarzschild solution is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 \left(d\theta^2 + \sin^2\theta d\phi^2\right), \tag{1.34}$$

or written in tensor form

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2M}{r}\right) & 0 & 0 & 0 \\ 0 & \left(1 - \frac{2M}{r}\right)^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2\theta \end{pmatrix}. \tag{1.35}$$

Here, M is the mass of our black hole. For comparison, the Minkowski metric of flat spacetime in spherical coordinates is

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2. \tag{1.36}$$

where $d\Omega^2 = \left(d\theta^2 + \sin^2\theta d\phi^2\right)$. An interesting property of the Schwarzschild spacetime is that it is actually the unique solution for any spherically symmetric vacuum spacetime. The static nature of Schwarzschild arises as a byproduct of the spherical symmetry condition.

Looking at the Schwarzschild solution, something interesting happens at $r = 2M$. The g_{rr} component blows up to infinity. When the metric components go to infinity, we call this a singularity. This radius is so important, that it's named the Schwarzschild radius,

$$R_S = 2M. \tag{1.37}$$

This radius defines the event horizon of the black hole. The event horizon is an invisible surface within which nothing (including light) can escape out to $r \geq 2M$.

Also, if $r = 0$, the g_{tt} component of the metric goes to infinity; another singularity. However, this singularity is different than R_S. We discussed in Sec. 1.2 the

importance of coordinate freedom and being able to switch coordinates. Let us make the following coordinate transformations

$$r^\star = r + 2M \ln\left(\frac{r}{2M} - 2\right), \tag{1.38}$$

$$v = t + r^\star, \tag{1.39}$$

$$u = t - r^\star. \tag{1.40}$$

With these transformations, our metric become

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dv^2 + (dvdr + drdv) + rd\Omega^2. \tag{1.41}$$

Now, we only have one singularity. Specifically when $r = 0$. Therefore, that means the $r = 2M$ singularity was what we call a "coordinate" singularity; i.e. not a "physical" singularity like $r = 0$. This is why we say that the event horizon is an invisible boundary because observers falling in to a black hole will be uninterrupted at $r = 2M$ and continue to fall. However, at $r = 0$, our classical theory of GR breaks down. In order to describe what happens at (or after) $r = 0$, we would need a valid theory of quantum gravity which we still do not yet have.

There is plenty more analysis one can do with the Schwarzschild solution to gain some intuition on how we should think of a curved spacetime. This includes computing the components of the Riemann tensor, Ricii tensor, and Christoffel symbols, computing the geodesics and stable orbits (Eq. 1.24), computing the Killing vectors, performing more coordinate transformations and expanding the Schwarzschild solution to negative radius, analyzing Kruskal diagrams, and more. All of which should give better intuition about the dynamics in a curved spacetime. There is also extension of the Schwarzschild solution to axial symmetry which is called the Kerr solution, again exploring all the same things for this new spacetime. For our purposes though, explaining R_S and the $r = 0$ physical singularity shall suffice.

1.4 Numerical Relativity

We now have a set of PDEs that we can solve to give us the metric at every point in our spacetime. Our job now is to put these equations onto the computer so we can solve them numerically. This is necessary because the Schwarzschild and Kerr solutions are the only analytic black hole solutions available, which means we will need numerical relativity (NR) to generate solutions of two black holes merging.

To put Einstein's equations onto the computer, we first need to split up the equations to somewhat disentangle the space and time dimensions from our spacetime. This

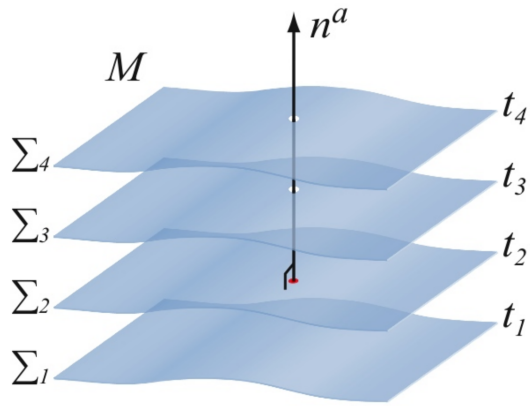

Figure 1.4: A foliation of spacetime into spatial hypersurfaces Σ_t, each labeled by a coordinate time t. The vector n^a is normal to each spatial hypersurface. This image was taken from [1].

procedure is called the "3+1" decomposition of spacetime. We do this by creating what is called a spacetime foliation. Fig. 1.4 depicts how we will foliate our spacetime. We start by splitting the spacetime up into spatial hypersurfaces Σ for every time t of our spacetime. Each Σ is a snapshot of full 3D space at every time (even though in the image they are represented as 2D slices, and time is the vertical axis). Then we can construct a vector n^a which is normal to our Σ hypersurfaces and points into the future. Note that for this section, we will switch from using greek letters (e.g. $\mu.\nu, \rho$) to latin letters (e.g. a, b, c) for spacetime symbols. This is purely a convention the NR community has adopted (albeit not a great one in our opinion).

We must be careful with our time coordinate. Recall from Sec. 1.2 that SR has no global time that applies to all spacetime points. This is still true in GR, so we must take this in to account. So instead of our time t being a single number, we will make it a function. We can use t write describe the normal vector n^a. We would like n_a to be normalized and timelike ($n^a n_a = -1$) so we define a normalization factor α to achieve this. Then, n^a becomes

$$n^a = -\frac{1}{\alpha}\nabla^a t. \tag{1.42}$$

Here, the lapse measures how much proper time has elapsed between spatial hypersurfaces.

Even though we now have how much time elapses between Σ, there's still more to our coordinates. Because of the coordinate freedom of GR, coordinates don't have to stay stationary between different Σ. In fact, we will show soon that we

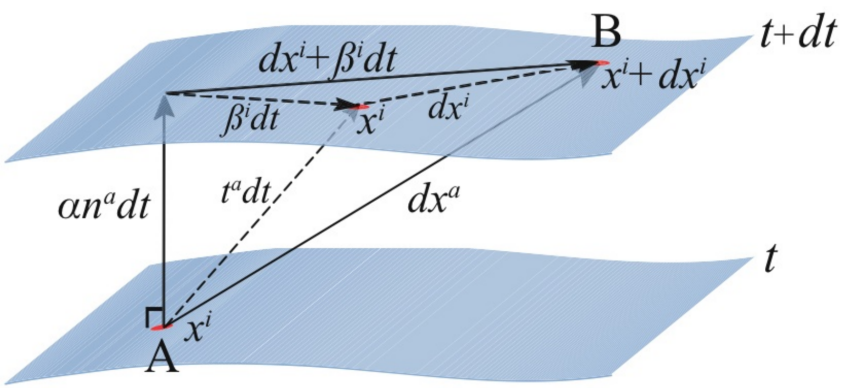

Figure 1.5: Illustration of the lapse α and shift vector β^i in the 3+1 decomposition of spacetime. The lapse α measures the proper time between adjacent hypersurfaces, while the shift β^i encodes how coordinates move between hypersurfaces. This image was taken from [1].

actually evolve the coordinates, implying that they will be different as time goes on. Our normal timelike normal vector n^a isn't enough to describe this movement. Therefore, we introduce a spatial vector $\beta^a = (0, \beta^i)$ called the shift which will tell us how much our coordinates have moved (or shifted) from the normal vector n^a. We can therefore construct a new vector t^a that will tell us how the coordinate that a grid point represents moves between spatial hypersurfaces

$$t^a = \alpha n^a + \beta^a. \tag{1.43}$$

Figure 1.5 is a great diagram that shows the breakdown of t^a into the timelike and spacelike components. It shows how those components map a spacetime coordinate at grid point A of one hypersurface to grid point B of the next hypersurface. Notice how αn^a gets us from the first hypersurface to the next, and β^i moves us within the next hypersurface. The lapse and shift together are four degrees of freedom, which is the exact number of coordinate degrees of freedom that GR lets us specify.

Now that we have the lapse and shift, we can move on to specifying our spacetime metric g_{ab} in terms of the lapse and shift. It will be very convenient to define a spatial metric within our spatial hypersurface Σ which we can use for quantities that live in Σ. We can combine our timelike normal vector with the spacetime metric to get

$$\gamma_{ab} = g_{ab} + n_a n_b. \tag{1.44}$$

There's a plus sign because n^a is defined with a minus sign. This means that γ_{ab} is purely spatial.

With this, we can represent our spacetime metric in terms of $\gamma_{ij}, \alpha, \beta^i$. If we choose $t^a = (1, 0, 0, 0)$, we can then compute n^a from Eq. 1.43 and solve Eq. 1.44 for g_{ab} as

$$g_{ab} = \begin{pmatrix} -\alpha^2 + \beta_l\beta^l & \beta_j \\ \beta_j & \gamma_{ij} \end{pmatrix}. \tag{1.45}$$

Or, expressed more commonly as a line element,

$$ds^2 \quad = g_{ab}dx^a dx^b = -\alpha^2 dt^2 + \gamma_{ij}\left(dx^i + \beta^i dt\right)\left(dx^j + \beta^j dt\right). \tag{1.46}$$

At this point, we are still only at the beginning of manipulating our equations. We would like to end up with equations that contain time derivatives ∂_t on the left hand side of the equation. These types of equations are ones that we can integrate in time to solve for our spacetime metric. We now branch off into explaining the two most popular formalisms for writing down evolution equations.

1.4.1 BSSN formulation

The first formalism we will discuss is the Baumgarte-Shapiro-Shibata-Nakamura (BSSN) formalism [2, 3]. First, we finish off defining the Arnowitt-Deser-Misner (ADM) formalism [4] as it's necessary for the BSSN formalism. This section is heavily based on derivations from [1].

We start by defining the extrinsic curvature K_{ab}. The extrinsic curvature is different than the Riemann curvature, or intrinsic curvature. The Riemann curvature tells us about how a spacetime itself is curved, and only relies on quantities defined in the spacetime. Said another way, intrinsic curvature does not rely on higher dimensions to define the curvature. However, our spatial slice Σ is a 3D space embedded in a higher dimensional (4D) space. Therefore, the extrinsic curvature will tell us how Σ is curved within the broader 4D spacetime. We define the extrinsic curvature by taking gradients of our normal vector n^a, and then projecting them onto Σ

$$K_{ab} = -\gamma_a{}^c\gamma_b{}^d\nabla_c n_d. \tag{1.47}$$

Using our definitions for γ_{ab}, we can rewrite this only in terms of n^a as

$$K_{ab} = -\nabla_a n_b - n_a n^c \nabla_c n_b. \tag{1.48}$$

We also introduce a new derivative operator, the Lie derivative $\mathcal{L}_{\mathbf{X}}$ as

$$\mathcal{L}_{\mathbf{X}} T^a{}_b = X^c\partial_c T^a{}_b - T^c{}_b\partial_c X^a + T^a{}_c\partial_b X^c. \tag{1.49}$$

With the Lie derivative, we can write the extrinsic curvature as

$$K_{ab} = -\frac{1}{2}\mathcal{L}_{\mathbf{n}}\gamma_{ab}. \tag{1.50}$$

We can also define the trace of the extrinsic curvature as

$$K = \gamma^{ab}K_{ab} = -\mathcal{L}_{\mathbf{n}} \ln \gamma^{1/2}. \tag{1.51}$$

To get time derivatives of the extrinsic curvature and our spatial metric, we will take the Lie derivative of each along our vector t^a. For the extrinsic curvature this will be (with a lot of simplification)

$$\begin{aligned}\mathcal{L}_{\mathbf{t}}K_{ab} &= \mathcal{L}_{\alpha\mathbf{n}+\beta}K_{ab} \\ &= \alpha\mathcal{L}_{\mathbf{n}}K_{ab} + \mathcal{L}_{\beta}K_{ab} \\ &= -D_aD_b\alpha + \alpha({}^{(3)}R_{ab} - 2K_{ac}K^c{}_b + KK_{ab}) + \mathcal{L}_{\beta}K_{ab}\end{aligned} \tag{1.52}$$

with ${}^{(3)}R_{ab}$ being the 3-dimensional Ricci tensor and $D_af = \gamma_a{}^b\nabla_b f$. For the spatial metric, this becomes

$$\begin{aligned}\mathcal{L}_{\mathbf{t}}\gamma_{ab} &= \mathcal{L}_{\alpha\mathbf{n}+\beta}\gamma_{ab} \\ &= \alpha\mathcal{L}_{\mathbf{n}}\gamma_{ab} + \mathcal{L}_{\beta}\gamma_{ab} \\ &= -2\alpha K_{ab} + \mathcal{L}_{\beta}\gamma_{ab}.\end{aligned} \tag{1.53}$$

Because of our definition of $t^a = (1, 0, 0, 0)$ and the fact that β is spatial, the following derivatives simplify to

$$\mathcal{L}_{\mathbf{t}} = \partial_t \tag{1.54}$$

$$\mathcal{L}_{\beta}K_{ij} = \beta^k D_k K_{ij} + K_{ik}D_j\beta^k + K_{kj}D_i\beta^k \tag{1.55}$$

$$\mathcal{L}_{\beta}\gamma_{ij} = D_i\beta_j + D_j\beta_i. \tag{1.56}$$

We could also take the trace of both the extrinsic curvature and spatial metric to get equations for the evolution of their traces

$$\partial_t \ln \gamma^{1/2} = -\alpha K + D_i\beta^i \tag{1.57}$$

$$\partial_t K = -D^2\alpha + \alpha K_{ij}K^{ij} + \beta^i D_i K. \tag{1.58}$$

There are actually extra terms in these equations for the stress-energy tensor following a similar procedure, but we exclude those here because we are only concerned with the vacuum equations.

We are unable to evolve the ADM equations stably [2, 3], so now we move on to the BSSN equations. We start by rewriting the spatial metric as

$$\gamma_{ij} = e^{4\phi}\bar{\gamma}_{ij} \tag{1.59}$$

where e^{ϕ} is a scalar function called the conformal factor, and $\bar{\gamma}_{ij}$ is called the conformal metric. This procedure of rewriting a quantity as a different quantity times a scalar function is a conformal transformation. The fourth power of e^{ϕ} is arbitrary, but useful for later calculation. The determinant of $\bar{\gamma}_{ij}$ is required to be $\bar{\gamma} = \eta = 1$. We then define the scalar function ϕ to be

$$\phi = \frac{1}{12}\ln(\gamma). \tag{1.60}$$

We also rewrite the extrinsic curvature as

$$K_{ij} = A_{ij} + \frac{1}{3}K\gamma_{ij} \tag{1.61}$$

where A_{ij} is the trace-free part of the extrinsic curvature. We also conformally transform this trace-free part as

$$A_{ij} = e^{4\phi}\bar{A}_{ij}. \tag{1.62}$$

Equation 1.56 for the time derivative of the spatial metric will now split into two equations, one for ϕ and one for $\bar{\gamma}_{ij}$ because of the conformal transformation. Those new equations are

$$\partial_t\phi = -\frac{1}{6}\alpha K^i_{\beta}\partial_i\phi + \frac{1}{6}\partial_i\beta^i \tag{1.63}$$

$$\partial_t\bar{\gamma}_{ij} = -2\alpha\bar{A}_{ij} + \beta^k\partial_k\bar{\gamma}_{ij} + \bar{\gamma}_{ik}\partial_j\beta^k + \bar{\gamma}_{kj}\partial_i\beta^k - \frac{2}{3}\bar{\gamma}_{ij}\partial_k\beta^k \tag{1.64}$$

In a similar fashion, Eq. 1.55 for the time derivative of the extrinsic curvature will also split into two equations, one for K and one for $\bar{A}_{ij}$ because of the conformal transformation. The new equations are

$$\partial_t K = -\gamma^{ij}D_jD_i\alpha + \alpha\left(\bar{A}_{ij}\bar{A}^{ij} + \frac{1}{3}K^2\right) + \beta^i\partial_i K, \tag{1.65}$$

$$\begin{aligned}\partial_t\bar{A}_{ij} = {}& e^{-4\phi}\big(-(D_iD_j\alpha)^{TF} + \alpha\,{}^{(3)}R^{TF}_{ij}\big) + \alpha\big(K\bar{A}_{ij} - 2\bar{A}_{il}\bar{A}^l_j\big)\\ &+ \beta^k\partial_k\bar{A}_{ij} + \bar{A}_{ik}\partial_j\beta^k + \bar{A}_{kj}\partial_i\beta^k - \frac{2}{3}\bar{A}_{ij}\partial_k\beta^k,\end{aligned} \tag{1.66}$$

where TF means the trace-free part of the tensor, e.g. ${}^{(3)}R^{TF}_{ij} = {}^{(3)}R_{ij} - {}^{(3)}R/3\gamma_{ij}$.

Next, we remember that with a conformal metric $\bar{\gamma}_{ij}$ there exists a conformal connection ${}^{(3)}\bar{\Gamma}^i_{jk}$ since connections are tied to metrics. We can contract the conformal connection to get

$$ {}^{(3)}\bar{\Gamma}^i \equiv \bar{\gamma}^{jk}\,{}^{(3)}\bar{\Gamma}^i_{jk} = -\partial_j \bar{\gamma}^{ij}. \tag{1.67} $$

In the BSSN formalism, we promote $\bar{\Gamma}^i$ from a computed quantity to an independent variable that is evolved along with all the others. If we take the time derivative of the previous equation, substitute in Eq. 1.64, and simplify a lot to get rid of the $\partial_j \bar{A}^{ij}$ term, we wind up with

$$ \begin{aligned} \partial_t^{(3)}\bar{\Gamma}^i = &-2\bar{A}^{ij}\partial_j\alpha + 2\alpha\left({}^{(3)}\bar{\Gamma}^i_{jk}\bar{A}^{kj} - \frac{2}{3}\bar{\gamma}^{ij}\partial_j K + 6\bar{A}^{ij}\partial_j\phi\right) \\ &+ \beta^j\partial_j^{(3)}\bar{\Gamma}^i - {}^{(3)}\bar{\Gamma}^j\partial_j\beta^i + \frac{2}{3}{}^{(3)}\bar{\Gamma}^i\partial_j\beta^j + \frac{1}{3}\bar{\gamma}^{li}\partial_l\partial_j\beta^j + \bar{\gamma}^{lj}\partial_j\partial_l\beta^i. \end{aligned} \tag{1.68} $$

Equations 1.63, 1.64, 1.65, 1.66, and 1.68 make up the evolution equations of the BSSN formalism. The last piece we'll need are evolution equations for the gauge functions α and β^i. Coming up with evolution equations for gauge quantities is more of an art than a science since, in theory, no matter what gauge we choose, we should still get the same physical system. However, solving equations numerically usually imposes more practical constraints.

There are several gauge conditions that have been tested and aren't popular choices to use with the BSSN equations. One is the geodesic slicing condition, $\alpha = 1$ and $\beta^i = 0$. Another is maximal slicing where $K = 0$ turns into an elliptic equation for the lapse, $D^2\alpha = \alpha K_{ij}K^{ij}$. Harmonic coordinates (which will be discussed more in Sec. 1.4.2) set $\Gamma^a = 0$.

Several of the mentioned conditions for the lapse can be described by

$$ (\partial_t - \beta^j\partial_j)\alpha = -\alpha^2 f(\alpha)K \tag{1.69} $$

where $f(\alpha)$ is a generic function. Choosing $f(\alpha) = 2/\alpha$, we can integrate Eq. 1.69 to get

$$ \alpha = 1 + \ln\gamma. \tag{1.70} $$

This is appropriately called the $1 + \log$ slicing condition and is very popular paired with the BSSN formalism.

A choice for the shift is to set $\partial_t\,{}^{(3)}\bar{\Gamma}^i = 0$. This is named "gamma freezing". Plugging this in to Eq. 1.68 will yield a very complicated equation to solve for

β^i. We can generalize and approximate this to something called a "gamma driver" equation

$$(\partial_t - \beta^j \partial_j)\beta^i = k(\partial_t \, {}^{(3)}\bar{\Gamma}^i + \frac{1}{2} \, {}^{(3)}\bar{\Gamma}^i), \tag{1.71}$$

where k is a positive constant. Since plugging in Eq. 1.68 still gives us a parabolic equation, we can make one more modification to get the hyperbolic gamma driver condition

$$(\partial_t - \beta^j \partial_j)\beta^i = \frac{3}{4} B^i \tag{1.72}$$

$$(\partial_t - \beta^j \partial_j)B^i = \partial_t \, {}^{(3)}\bar{\Gamma}^i - \frac{1}{2} B^i. \tag{1.73}$$

The $1 + \log$ slicing for the lapse and the both gamma driver equations are the most popular gauge conditions to choose for the BSSN formalism. Small tweaks to each of these exist in the different codes that use the BSSN formalism. One such tweak is removing the advective term $\beta^j \partial_j$ from the left hand side of the equations. These two conditions combined are referred as the "moving puncture" conditions since they follow where the singularities of each black hole would be.

1.4.2 Generalized Harmonic formulation

The next (and final) formalism we will discuss is the generalized harmonic (GH) formulation [5]. This differs quite a bit from the BSSN formalism, but we will still borrow some things as we need them. Namely, we will still keep the standard 3+1 decomposition of the metric since having the lapse, shift, and spatial metric will be needed.

$$ds^2 \quad = g_{ab} dx^a dx^b = -\alpha^2 dt^2 + \gamma_{ij} \left(dx^i + \beta^i dt\right)\left(dx^j + \beta^j dt\right). \tag{1.74}$$

Right away, we will start with the gauge as this formalism is based around using harmonic coordinates. Given coordinates x^a where each x^a is treated as a scalar function, we have the relation that

$$\begin{aligned} \Box x^a = g^{bc} \nabla_b \nabla_c x^a &= g^{bc} \nabla_b (\partial_c x^a) \\ &= g^{bc} (\partial_b \partial_c x^a - \Gamma^d_{bc} \partial_d x^a) \\ &= -g^{bc} \Gamma^a_{bc} \\ &= -\Gamma^a \end{aligned} \tag{1.75}$$

$\Box$ is the d'Alembertian operator and we have used the fact that since x^a is a scalar and are coordinates, $\nabla_c x^a = \partial_c x^a = \delta^a_c$. We then express the harmonic gauge

condition as

$$\Box x^a = -\Gamma^a = 0. \tag{1.76}$$

The generalized harmonic gauge condition is then related to the harmonic gauge condition, but instead of setting $-\Gamma^a = 0$, we set it as

$$\Box x^a = -\Gamma^a = H^a(x, g), \tag{1.77}$$

where H^a is some arbitrary function of the coordinates and spacetime metric (but not derivatives of the metric!) that we specify. These conditions on H^a are critical because they preserve the strong hyperbolicity of the Einstein equations, which allow for stable evolutions of spacetime. If these conditions were not met, any tiny instability or error in our evolutions would grow exponentially and spoil our solution. And since we are using computers to solve these problems, the finite truncation of floating point numbers will always introduce some tiny error so we can be guaranteed an unstable evolution.

Now in the BSSN formalism, we showed that the lapse and the shift contain our coordinate freedom and to impose a gauge condition, we had to specify the lapse and shift either analytically or by evolving them alongside the spacetime quantities. Here we show that specifying H^a is equivalent to specifying the lapse and shift with the following evolution equations[5]

$$\partial_t \alpha - \beta^j \partial_j \alpha = -\alpha(H_0 - \beta^i H_i + \alpha K) \tag{1.78}$$

$$\partial_t \beta^i - \beta^j \partial_j \beta^i = \alpha \gamma^{ij} \left[\alpha(H_j + \gamma^{kl}\Gamma_{jkl}) - \partial_j \alpha\right] \tag{1.79}$$

where K is the extrinsic curvature and $\gamma^{ij}\Gamma_{jkl} = \Gamma^i_{kl}$. Here we can see that even though it may not be easy to convert between the lapse and shift and H^a, specifying one or the other is equivalent.

In the GH formulation, we won't be evolving quantities like γ_{ij}, K_{ij}, or Γ^i. Instead we will be evolving the full spacetime metric g_{ab} itself. The specific form of the GH formulation that we will be presenting is the first order symmetric hyperbolic GH formulation. In the typical formulation, we would time integrate the second time derivative ∂_t^2 of the metric. In the first order formulation, we only evolve ∂_t. This is advantageous because there are a number of well-known standard integrators that work on first order hyperbolic equations. However, this requires us to some how reduce the order of the PDEs since Einstein's equations are second order.

To show the procedure, we first start with an example of reducing the order of a scalar wave function

$$\Box\psi = \left(\frac{\partial^2}{\partial t^2} - \nabla^2\right)\psi = 0. \tag{1.80}$$

We choose this equation as an example because Einstein's equations themselves are basically really complicated wave equations due to the of the second order derivatives in both space and time that appear from the Riemann tensor. To reduce the order of the scalar wave system, we introduce two new auxiliary variables

$$\Pi \equiv -\partial_t\psi, \tag{1.81}$$

$$\Phi_i \equiv \partial_i\psi. \tag{1.82}$$

Using these new variables, we can take time derivatives of each to get a system of equations equivalent to Eq. 1.80

$$\partial_t\psi = -\Pi \tag{1.83}$$

$$\partial_t\Pi = -\partial^i\Phi_i \tag{1.84}$$

$$\partial_t\Phi_i = -\partial_i\Pi \tag{1.85}$$

We have transformed a second order equation for ψ into a system of first order equations for ψ, Π, Φ_i. It can be useful to combine these evolved variables in a single vector $u^A = \{\psi, \Pi, \Phi_i\}$. We can then rewrite the evolution equations as

$$\partial_t u^A + M^j\partial_j u^A = S(u^A) \tag{1.86}$$

where M^j is a coefficient matrix and $S(u^A)$ is a source vector that does not contain any derivatives of u^A. In this form, if all the left-eigenvectors of M^j are linearly independent, then the system is said to be strongly hyperbolic. We will return to this shortly.

Notice that Eq. 1.81 for Π became an evolution equation, but Eq. 1.82 for Φ_i isn't an evolution equation because it doesn't have time derivatives. It is actually called a constraint equation because now we evolve ψ and Φ_i independently. We can define two constraint with Φ_i

$$\mathcal{C}_i = \partial_i\psi - \Phi_i = 0 \tag{1.87}$$

$$\mathcal{C}_{ij} = \partial_i\Phi_j - \partial_j\Phi_i = 0 \tag{1.88}$$

Therefore, we can actually use $\mathcal{C}_i$ and $\mathcal{C}_{ij}$ throughout the simulation to check if our evolution is consistent. Additionally, and more importantly, the version of the

evolution equations above will leave errors around in our solution. To mitigate this, we can actually alter the third equation to depend on $\mathcal{C}_i$ which will damp away any errors that arise. We do this by adding a multiple of $\mathcal{C}_i$ to the Φ_i equation. The modified equations are

$$\partial_t \psi = -\Pi \tag{1.89}$$

$$\partial_t \Pi = -\partial^i \Phi_i \tag{1.90}$$

$$\partial_t \Phi_i = -\partial_i \Pi + \gamma_2(\partial_i \psi - \Phi_i) \tag{1.91}$$

where γ_2 is some analytic function specified for all space and time.

With these new equations we have that

$$M^j = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ -\gamma_2 & 1 & 0 \end{pmatrix}, \quad S(u^A) = \begin{pmatrix} -\Pi \\ 0 \\ -\gamma_2 \Phi_i \end{pmatrix}. \tag{1.92}$$

The left eigenvectors, or characteristic fields, of this system are

$$w^\psi = \psi \tag{1.93}$$

$$w_i^0 = (\delta_i^k - n^k n_i)\Phi_k \tag{1.94}$$

$$w^\pm = \Pi \pm n^k \Phi_k - \gamma_2 \psi \tag{1.95}$$

where n^k is a vector in whatever direction we are interested in analyzing. The eigenvalues, or characteristic speeds, associated with these eigenvectors are

$$\lambda_\psi = 0 \tag{1.96}$$

$$\lambda_0 = 0 \tag{1.97}$$

$$\lambda_\pm = \pm 1 \tag{1.98}$$

And while we won't discuss their use here, it is also useful to express our evolved variables u^A in terms of the characteristic fields are

$$\psi = w^\psi \tag{1.99}$$

$$\Pi = \frac{1}{2}(w^+ + w^-) + \gamma_2 w^\psi \tag{1.100}$$

$$\Phi = \frac{1}{2}(w^+ - w^-) + w_i^0 \tag{1.101}$$

Lastly, we take a time derivative of our constraint $\mathcal{C}_i$ to see how it evolves in time.

$$\partial_t \mathcal{C}_i = \partial_t \partial_i \psi - \partial_t \Phi_i \tag{1.102}$$

$$= -\partial_i \Pi - (-\partial_i \Pi + \gamma_2 \mathcal{C}_i) \tag{1.103}$$

$$= -\gamma_2 \mathcal{C}_i \tag{1.104}$$

where we used $\partial_t \psi = \Pi$ and Eq. 1.91. The solution to this differential equation is $\propto e^{-\gamma_2 t}$, so our constraints will be exponentially damped. A similar equation exists for $\mathcal{C}_{ij}$

$$\partial_t \mathcal{C}_{ij} = -\gamma_2 \mathcal{C}_{ij}. \tag{1.105}$$

With these characteristic fields, speeds, and constraints, one is also able to impose boundary conditions on the incoming characteristic fields rather than evolved variables themselves. However, here we skip that as it isn't as informative.

Now we can return back to the GH formulation. For gravity, our field will be the metric itself g_{ab} instead of ψ. Then we define the two auxiliary variables as

$$\Pi_{ab} = -n^c \partial_c g_{ab} = -\frac{1}{\alpha}\left(\partial_t g_{ab} - \beta^i \partial_i g_{ab}\right) \tag{1.106}$$

$$\Phi_{iab} = \partial_i g_{ab} \tag{1.107}$$

where $n^c = (\alpha^{-1}, -\alpha^{-1}\beta^i)$ is a timelike unit normal vector, normalized with the spacetime metric in Eq. 1.46. The equation for Φ_{iab} and our gauge condition in Eq.1.77 give us two constraints

$$\mathcal{C}_a = H_a + \Gamma_a = 0 \tag{1.108}$$

$$\mathcal{C}_{iab} = \partial_i g_{ab} - \Phi_{iab} = 0. \tag{1.109}$$

Rearranging Eq. 1.106, we can get

$$\partial_t g_{ab} - \beta^i \partial_j g_{ab} = -\alpha \Pi_{ab}. \tag{1.110}$$

And if we take the partial derivative of this and rearrange, we can get an equation for $\partial_t \Phi_{iab}$. To account for constraint damping, we add $\gamma_1 \beta^i \mathcal{C}_{iab}$ to the equation for $\partial_t g_{ab}$ and $\gamma_2 \alpha \mathcal{C}_{iab}$ to the equation for $\partial_t \Phi_{iab}$. The equation for $\partial_t \Pi$ is unfortunately somewhat complicated to derive so we omit its derivation, but it will contain terms with $\mathcal{C}_a$ and $\gamma_1 \gamma_2 \beta^i \mathcal{C}_{iab}$ for constraint damping. Instead, we simply present the

first-order symmetric hyperbolic GH system as

$$\partial_t g_{ab} = (1+\gamma_1)\,\beta^k \partial_k g_{ab} - \alpha \Pi_{ab} - \gamma_1 \beta^i \Phi_{iab}, \tag{1.111}$$

$$\begin{aligned}\partial_t \Phi_{iab} &= \beta^k \partial_k \Phi_{iab} - \alpha \partial_i \Pi_{ab} + \alpha\gamma_2 \partial_i g_{ab} + \frac{1}{2}\alpha n^c n^d \Phi_{icd}\Pi_{ab} \\ &\quad + \alpha \gamma^{jk} n^c \Phi_{ijc}\Phi_{kab} - \alpha\gamma_2 \Phi_{iab},\end{aligned} \tag{1.112}$$

$$\begin{aligned}\partial_t \Pi_{ab} &= \beta^k \partial_k \Pi_{ab} - \alpha\gamma^{ki}\partial_k \Phi_{iab} + \gamma_1\gamma_2 \beta^k \partial_k g_{ab} \\ &\quad + 2\alpha g^{cd}\left(\gamma^{ij}\Phi_{ica}\Phi_{jdb} - \Pi_{ca}\Pi_{db} - g^{ef}\Gamma_{ace}\Gamma_{bdf}\right) \\ &\quad - 2\alpha \nabla_{(a}H_{b)} - \frac{1}{2}\alpha n^c n^d \Pi_{cd}\Pi_{ab} - \alpha n^c \Pi_{ci}\gamma^{ij}\Phi_{jab} \\ &\quad + \alpha\gamma_0\left(2\delta^c{}_{(a}n_{b)} - g_{ab}n^c\right)\mathcal{C}_c - \gamma_1\gamma_2\beta^i\Phi_{iab}.\end{aligned} \tag{1.113}$$

Just like the scalar wave system, we can compute the characteristic fields as

$$w^g_{ab} = g_{ab}, \tag{1.114}$$

$$w^0_{iab} = (\delta^k_i - s^k s_i)\Phi_{kab}, \tag{1.115}$$

$$w^\pm_{ab} = \Pi_{ab} \pm s^i \Phi_{iab} - \gamma_2 g_{ab}, \tag{1.116}$$

along with the characteristic speeds

$$\lambda_{w^g} = -\,(1+\gamma_1)\beta^i s_i, \tag{1.117}$$

$$\lambda_{w^0} = -\,\beta^i s_i, \tag{1.118}$$

$$\lambda_{w^\pm} = \pm\,\alpha - \beta^i s_i, \tag{1.119}$$

where s^i is the unit normal vector to a 2D surface, the analog of n^k for the scalar wave system. Notice that if $\beta^i s_i = 0$, these characteristic speeds are strikingly similar to the ones in Eqs. 1.96-1.98 for the scalar wave system.

The evolved variables in terms of the characteristic speeds are then

$$g_{ab} = w^g_{ab}, \tag{1.120}$$

$$\Pi_{ab} = \frac{1}{2}\left(w^+_{ab} + w^-_{ab}\right) + \gamma_2 w^g_{ab}, \tag{1.121}$$

$$\Phi_{iab} = \frac{1}{2}\left(w^+_{ab} - w^-_{ab}\right) s_i + w^0_{iab}. \tag{1.122}$$

Again, these are very similar to the scalar wave system, just with the extra spacetime indices.

In addition to $\mathcal{C}_a$ and $\mathcal{C}_{iab}$, we are able to define more constraints $\mathcal{C}_{ia}$, $\mathcal{C}_{ijab}$, and $\mathcal{F}_a$. We don't define them here because only $\mathcal{C}_a$ and $\mathcal{C}_{iab}$ are added to the evolution

equations to damp away errors. However, they are still useful for monitoring the simulation for errors. We can also combine all of these constraints into a single quantity called the "constraint energy" which gives us an overall notion of how much energy is contained in constraints,

$$\begin{aligned}\mathcal{E} = \delta^{ab}\left[\mathcal{C}_a\mathcal{C}_b + \left(\mathcal{F}_a\mathcal{F}_b + \mathcal{C}_{ia}\mathcal{C}_{jb}\gamma^{ij}\right)\right] \\ + \delta^{ab}\delta^{cd}\left(\mathcal{C}_{iac}\mathcal{C}_{jbd}\gamma^{ij} + \mathcal{C}_{ikac}\mathcal{C}_{jlbd}\gamma^{ij}\gamma^{kl}\right).\end{aligned} \tag{1.123}$$

Ideally, we'd like zero constraint energy, but numerical error means we must settle for this being as small as possible. Again, we don't go over boundary conditions here, but refer the reader to [5] and [6] for more details.

At this point we will stop our overview of NR. There are still a number of different topics which we did not cover such as (a) initial data on the first spatial hypersurface, (b) boundary conditions for either BSSN or GH, (c) GW extraction via finite radius, extrapolation, or CCE (see Chapter **??**), (d) numerical methods for how to discretize the mesh, (e) methods for taking "good" numerical derivatives, (f) finding the horizon of a black hole (see Sec. [TODO:CITE]), or (g) any of the previous topics applied to simulating matter around black holes or neutron stars. This field is a rich mix of theory and computational algorithms and deserves to be explored further.

1.5 Goals of NR

Since we now have an idea of how to carry out numerical simulations of black holes, we try to motivate the *why* of it all and what the purpose of NR is. Einstein predicted GWs back in the early 20th century [7, 8]. However, the predictions on the amplitude of the GWs were so small, $\Delta L \sim 10^{-21}$, that they were thought to be impossible to measure. Technology eventually advanced sufficiently far that we are now able to measure these GWs here on Earth using lasers in 4km long vacuum tubes. The first ever signal of two black holes merging was detected by the LIGO [9], VIRGO [10], and KAGRA [11] collaborations in 2015 [12, 13].

But when LIGO detected this merger, how did it know it was a merger of two black holes? How did it know that it was a merger at all? Well there had to be some prediction about what the GWs would look like so data analysts would be able to search the observed data for the predicted signal. These predictions must be based on some sort of theory.

Because data analysts don't know what parameters black holes will have when they see a potential signal, our theory must be able to give a signal with any BH

parameters very quickly. Actually solving Einstein's equations numerically can be very expensive, so a perturbative theory like Post-Newtonian[TODO:CITE]or Post-Minkowskian[TODO:CITE]is usually better for quick evaluation.

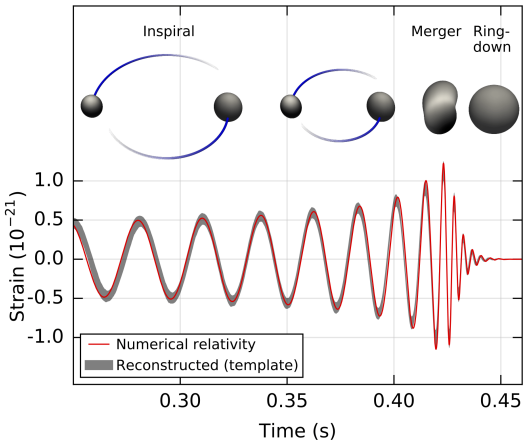

Figure 1.6: The GW150914 signal showing the inspiral, merger, and ringdown of the BBH. Figure taken from [12].

However, these perturbative theories break down when the black holes are close to merger because the perturbations of the background spacetime become too large. The perturbative methods are useful during the inspiral. This is precisely the role that numerical relativity plays in providing a template bank of waveforms for LIGO data analysis. NR simulations provide accurate waveforms during the plunge, merger, and ringdown phase of merger. Figure 1.6 shows the phases of inspiral, merger, and ringdown. Plunge happens just before merger.

1. Ultimate goals of Numerical Relativity (models/training)
2. Why HPC to achieve goals
3. GH and why horizon finding is important
4. Chapters of thesis
    a) Brief on Lovelace and Nelli et al.
    b) Brief on Nelli et al.
    c) Brief on Cross code comparison

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